

Latest on the Block

Small inverters for the masses

Manufacturers supply inverters and UPS in different sizes for various purposes. The price of a unit depends upon its output capacity, the battery backup and reputation of the brand.

Due to the increase in power cuts, customers are placing more number of orders. Generally, people prefer inverters that are inexpensive, safe, and produce no noise and pollution. So companies are planning to manufacture small inverters and home UPS that are affordable for the general public. They are trying to provide low-cost solutions by resorting to techniques such as the use of aluminium wire in transformers, which reduces the cost substantially.

Solar micro inverters

With the consumer looking for greener and safer technology, manufacturers such as Convergence Power Systems, Delta Energy Systems, PEPTELLER, Su-Kam, Truepower International and Upline Energy Systems are introducing an array of solar inverters.

Solar inverters are giving stiff competition to their conventional counterparts. These inverters find applications

in household and solar installations (suitable for all household electrical appliances such as computers, televisions and microwaves), rural electrification, commercial/industrial systems and utility-scale PV power generation plants.

A conventional inverter and a solar one don't differ much, except for the energy supply. The conventional inverter converts DC from a battery into AC. A solar inverter, on the other hand, uses a string of PV panels as the DC source. Power inverters use AC mains to charge batteries, whereas solar inverters use pure solar energy.

Among the developing technologies, solar hybrid inverters are one of the preferred options. These inverters change the DC from the PV array and battery into AC for use with electrical appliances. If the mains and solar power are both available, the charging current is taken from both sources, with priority given to solar power.

Micro inverters using solar energy are a recent phenomenon. These systems have started replacing solar string inverters in small residential solar installations. The difference between a solar micro inverter and a solar inverter is that the former works with a single solar

panel, the latter with 10 to 50,000 solar panels, depending on its size.

Solar micro inverters are more reliable as the failure of a single piece affects only that solar panel and not the entire array of solar panels, as in the case of solar inverters.

Inverters for farmers

In 40 per cent of electrified rural Indian households, the average power cuts last 14-16 hours a day, causing problems to millions of farmer families. So they need a back-up solution that is efficient enough to power a household's electrical appliances and at the same time be economical enough to fit the farmer's spending capacity.

Su-kam has developed the Kisan Inverter that can be connected to the existing automotive battery of a tractor when the tractor is idle, allowing the users to run most household appliances like tubelights, fans and TV. With the Kisan Inverter, the users need not spend money to buy a separate inverter battery. A specially designed software controls the battery cut-off, ensuring that the battery is not totally discharged. ●

The author is executive editor at EFY

